

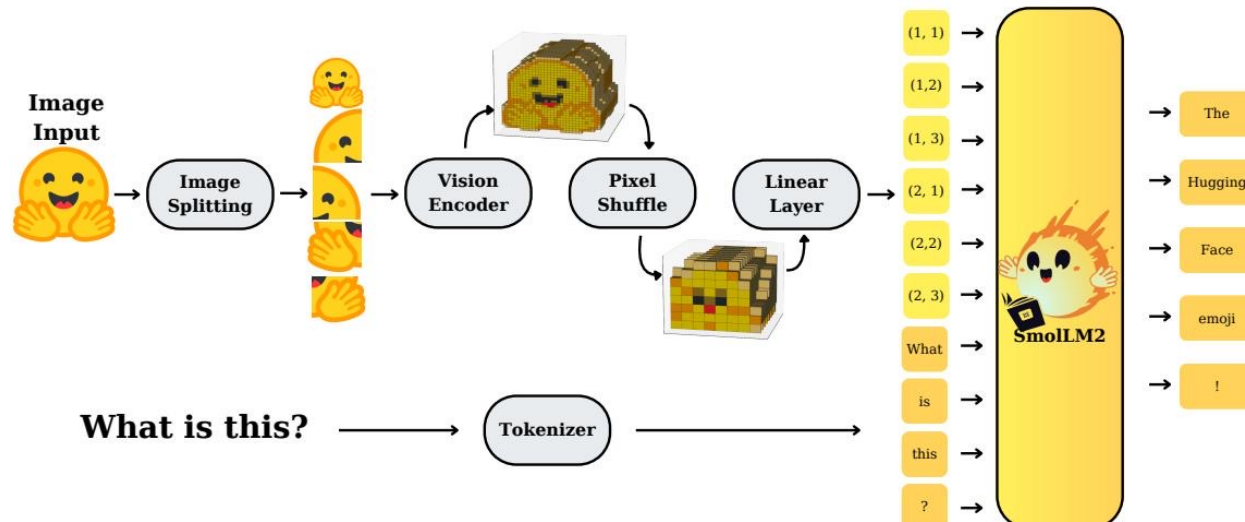
SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University
Compact
vision-language models for edge devices

Large VLMs Are Powerful but Impractical for Edge Deployment

- The Problem: State-of-the-art Vision-Language Models (VLMs) require massive compute and memory.
- Edge Limitations: Mobile devices and edge hardware cannot support models demanding tens of gigabytes of RAM.
- Example: Molmo-A1B-7B requires 27.7 GB of RAM, rendering it unusable on standard edge devices.
- The Solution: SmolVLM achieves sub-1GB RAM usage, enabling true on-device processing without cloud dependency.

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone



Pipeline Flow:

1. Image Splitting: High-resolution images are split into sub-images.
2. Vision Encoder: Processed by a compact SigLIP vision encoder.
3. Pixel Shuffle: Compresses features via space-to-depth operations.
4. Projection & LM: Linearly projected into the SmolLM2 language model and concatenated with text tokens.

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: Balanced encoder-LM compute allocation is critical. When paired with a 135M language model, a 93M SigLIP-B/16 outperforms a larger 428M SigLIP-SO400M. The larger encoder only justifies its size at the 1.7B LM scale.



Superior

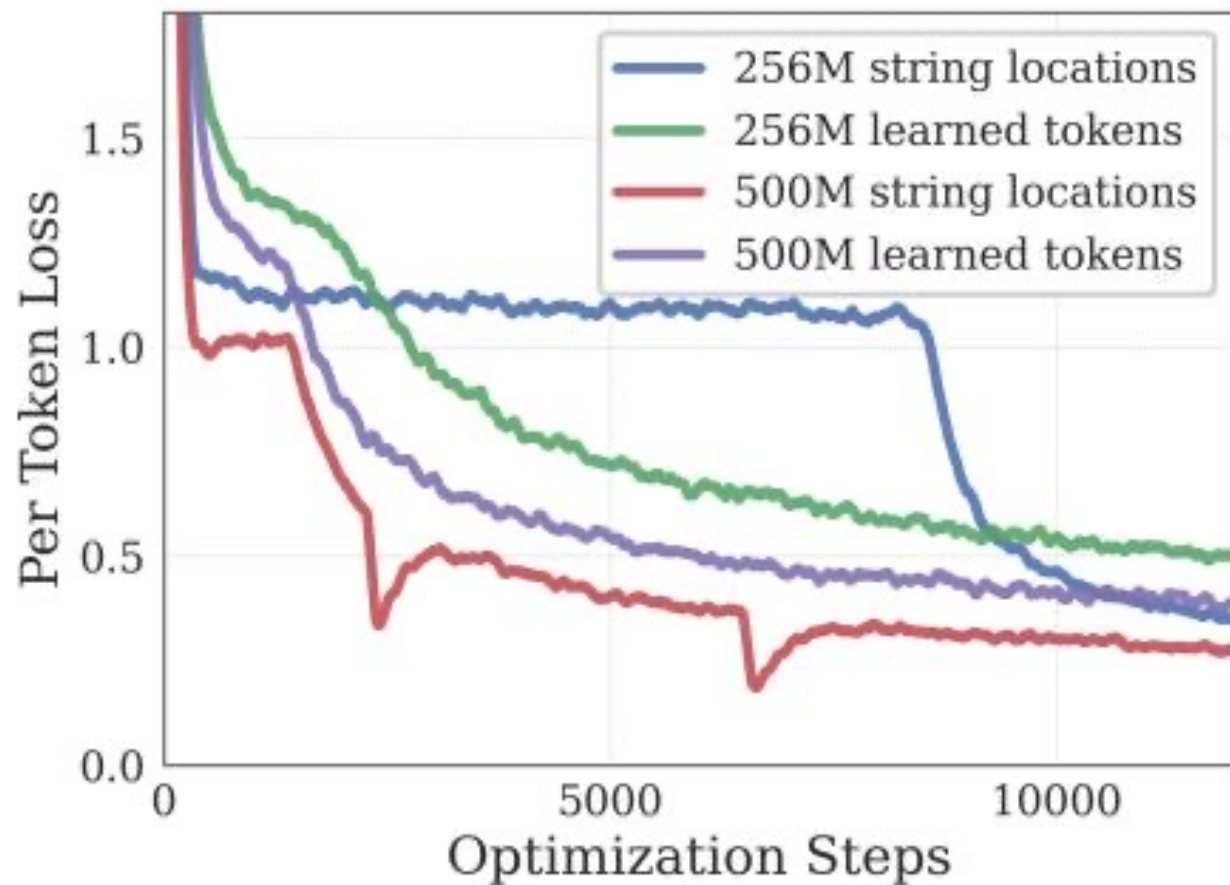
Performance
Vision Encoder: 93M SigLIP-B/16 + Language Model: 135M SmolLM2



Inferior

Performance
Vision Encoder: 428M SigLIP-SO400M + Language Model: 135M SmolLM2

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models



Finding 3: Aggressive Token Compression

- Pixel shuffle rearranges spatial features into channels (space-to-depth).
- Reduces the number of visual tokens by a factor of r^2 .
- While larger models use $r=2$ (4×), small VLMs benefit from $r=4$ (16×).
- The 16× reduction significantly eases attention overhead and improves long-context modeling performance.

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2: Extended Context Length

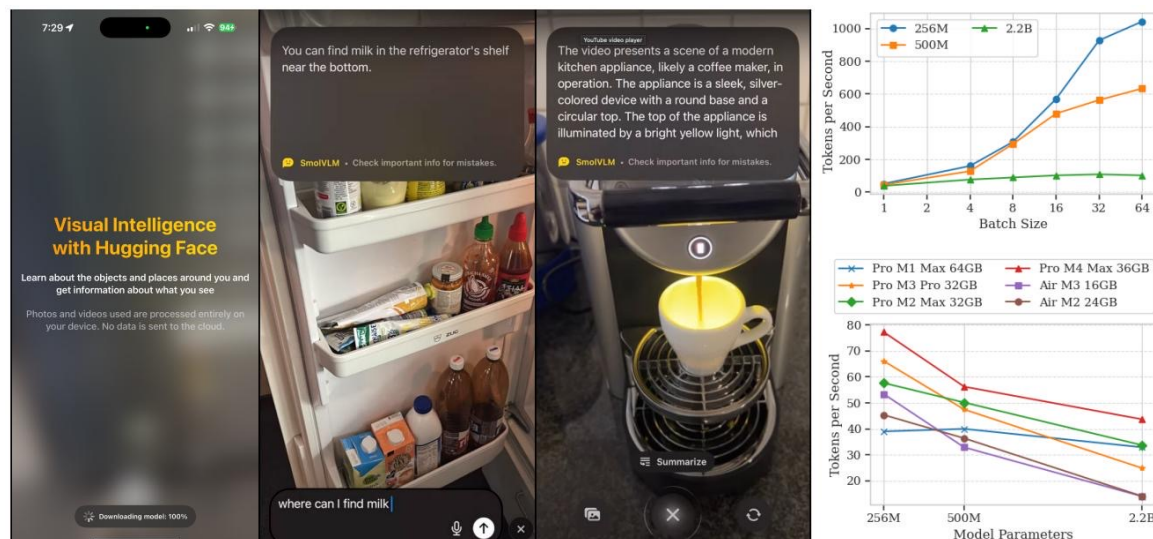
- Performance improves significantly when extending context length from 2K to 16K tokens.
- Methodology: Increased the RoPE base from 10K to 273K and fine-tuned on mixed long/short-context data.
- SmolVLM 2.2B utilizes the full 16K context window.
- SmolVLM 256M and 500M variants use 8K context windows.
- Impact: Enables robust long-context understanding and processing of higher resolution images without truncation.

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Benchmark	SmolVLM 256M\n(0.8 GB)	SmolVLM 500M\n(1.2 GB)	SmolVLM 2.2B\n(4.9 GB)	Best Efficient OS\n(Molmo/InternVL2)
OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo-A1B-7B)
AI2D	46.4%	59.2%	70.0%	71.0% (Molmo-A1B-7B)
ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo-A1B-7B)
TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo-A1B-7B)
DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo-A1B-7B)
ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo-A1B-7B)
MMMU (Multi)	29.0%	33.7%	42.0%	33.9% (Molmo-A1B-7B)
MathVista	35.9%	40.1%	51.5%	37.6% (Molmo-A1B-7B)
MMStar	34.6%	38.3%	46.0%	43.1% (Molmo-A1B-7B)
Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2-2B)
MLVU (Video)	40.6%	47.3%	55.2%	48.2% (InternVL2-2B)
Average Score	44.0%	51.0%	59.8%	—
RAM Usage	0.8 GB	1.2 GB	4.9 GB	27.7 GB (Molmo-A1B-7B)

* SmolVLM-256M surpasses the 300× larger Idefics-80B. RAM usage shown at batch=1.

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon



On-Device Performance:

- Local Execution: Runs entirely locally via the HuggingSnap iOS app, no cloud dependency.
- Pro M1 Max 64GB: Achieves up to ~80 tokens/second using SmolVLM-256M.
- High-End Devices: At batch size 64, the 256M model reaches nearly 1000 tokens/second.
- MacBook Air M3 16GB: Delivers ~55 tokens/second for SmolVLM-256M.
- Scalability: Throughput scales roughly linearly with batch size for the smallest

Key Takeaways: Design Principles for Efficient Multimodal Models

1. Balance Encoder-LM Compute: Smaller vision encoders (e.g., SigLIP-B/16) are optimal for compact LMs.
2. Extend Context Length: Scaling context up to 16K unlocks higher image resolution handling.
3. Aggressive Token Compression: A 16× reduction (r=4 pixel shuffle) optimizes attention overhead for smaller models.
- 4. Sub-1GB Inference is Achievable: Architectural efficiency outweighs sheer scale, enabling powerful capabilities on smartphones and edge devices.**
5. Fully Open-Source: All weights, datasets, code, and the HuggingSnap mobile app are open-sourced.